MI-06: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. Detailed review and scope.

1 MI-06: no finite two-unitary constant for the arithmetic symmetric modulus

Define the arithmetic symmetric modulus by

$$\mathcal{S}(X) = \frac{|X| + |X^*|}{2}.$$

The target proposes that for every X, Y there are unitaries U, V with

$$\mathcal{S}(X+Y) \le \sqrt{2}\big(U\mathcal{S}(X)U^* + V\mathcal{S}(Y)V^*\big).$$

The following result disproves that assertion and the more general finite-constant question in [1, Conjecture 1.6 and equation (1.4)].

Theorem 1.1. *Even in dimension three, there is no finite nonnegative constant C such that every pair $X, Y \in \mathbb{M}_3(\mathbb{C})$ admits unitaries U, V satisfying*

$$\mathcal{S}(X+Y) \le C\big(U\mathcal{S}(X)U^* + V\mathcal{S}(Y)V^*\big). \tag{1}$$

The failure is witnessed by real rank-one matrices.

Proof. For $t > 0$, let

$$X = e_1(e_1 + te_2)^* = \begin{pmatrix} 1 & t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad Y = -(e_1 + te_3)e_1^* = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ -t & 0 & 0 \end{pmatrix}.$$

Put $s = \sqrt{1+t^2}$. If $Z = uv^*$ has rank one, then

$$|Z| = \frac{\|u\|}{\|v\|} vv^*, \qquad |Z^*| = \frac{\|v\|}{\|u\|} uu^*.$$

Applying this identity shows that both $\mathcal{S}(X)$ and $\mathcal{S}(Y)$ have eigenvalues

$$a = \frac{s+1}{2}, \qquad b = \frac{s-1}{2}, \qquad 0.$$

On the other hand,

$$X + Y = t(e_1e_2^* - e_3e_1^*)$$

has right and left moduli $t\,\mathrm{diag}(1,1,0)$ and $t\,\mathrm{diag}(1,0,1)$, respectively. Hence

$$\mathcal{S}(X+Y) = \mathrm{diag}(t, t/2, t/2) \geq (t/2)I_3. \tag{2}$$

Fix arbitrary unitaries U, V. Choose a unit vector w orthogonal to a top eigenvector of $U\mathcal{S}(X)U^*$ and to a top eigenvector of $V\mathcal{S}(Y)V^*$. Such a vector exists because the ambient dimension is three. The eigenvalue lists above imply

$$w^*\big(U\mathcal{S}(X)U^* + V\mathcal{S}(Y)V^*\big)w \leq 2b = s-1.$$

Combining this with (2), any domination (1) would force

$$\frac{t}{2} \leq C(s-1), \qquad C \geq \frac{t}{2(\sqrt{1+t^2}-1)} = \frac{\sqrt{1+t^2}+1}{2t}.$$

The final expression tends to $+\infty$ as $t \downarrow 0$. □

Corollary 1.2 (A small exact counterexample to $\sqrt{2}$)**.** *Take $t = 3/4$ in the preceding construction. Then*

$$\mathrm{spec}\,\mathcal{S}(X) = \mathrm{spec}\,\mathcal{S}(Y) = \{9/8, 1/8, 0\}, \qquad \mathcal{S}(X+Y) = \mathrm{diag}(3/4, 3/8, 3/8).$$

For arbitrary U, V, the quadratic form of the proposed right-hand side on the vector w used above is at most $\sqrt{2}/4 < 3/8$. Thus the conjectured domination fails.

For direct verification, the two individual symmetric moduli in this example are rational matrices:

$$\mathcal{S}(X) = \begin{pmatrix} 41/40 & 3/10 & 0 \\ 3/10 & 9/40 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathcal{S}(Y) = \begin{pmatrix} 41/40 & 0 & 3/10 \\ 0 & 0 & 0 \\ 3/10 & 0 & 9/40 \end{pmatrix}.$$

This obstruction concerns order domination by two separately rotated summands. It does not contradict a unitarily invariant norm inequality for their sum, and it does not identify the arithmetic modulus with the quadratic symmetric modulus.

References

[1] Teng Zhang. Operator symmetric moduli and sharp triangle inequalities. arXiv:2603.01046, 2026. Version consulted: v1; Conjecture 1.6, equation (1.4), Problems 1.11 and 1.17. URL: https://arxiv.org/abs/2603.01046.